

Project Documentation

Macroeconomic Analysis of Consumption Expenditure

1. Project Overview

This project delivers a comprehensive, data-driven macroeconomic analysis of private final consumption expenditure in the domestic market across a seven-year timeline spanning 2004-05 to 2010-11. By executing high-fidelity data transformations on extensive national accounting datasets, the primary goal was to structurally isolate core consumption pillars, uncover long-term compound growth patterns, identify shifting consumer behavior, and visually communicate strategic market insights through an interactive business intelligence framework.

2. Tools Used

- **Microsoft Excel:** Deployed as the baseline engine for staging raw micro-records, parsing structural data matrices, cleaning null and nested entities, and verifying mathematical reconciliation via multi-axis pivot structures.
- **Power BI Desktop:** Utilized to construct a dynamic star-schema data model, engineer calculated columns, formulate time-series metrics using DAX (Data Analysis Expressions), and engineer a production-ready analytical dashboard.

3. Dataset

- **Source:** National Accounts Historical Economic Statistics and Consumption Data Files (2004–2011).
- **Data Footprint:** The database tracks historical records split across multiple granular columns including Item Name, Parent Group Name, and chronological spending values mapping out seven fiscal cycles: 2004-05, 2005-06, 2006-07, 2007-08, 2008-09, 2009-10, and 2010-11.

4. Steps Followed

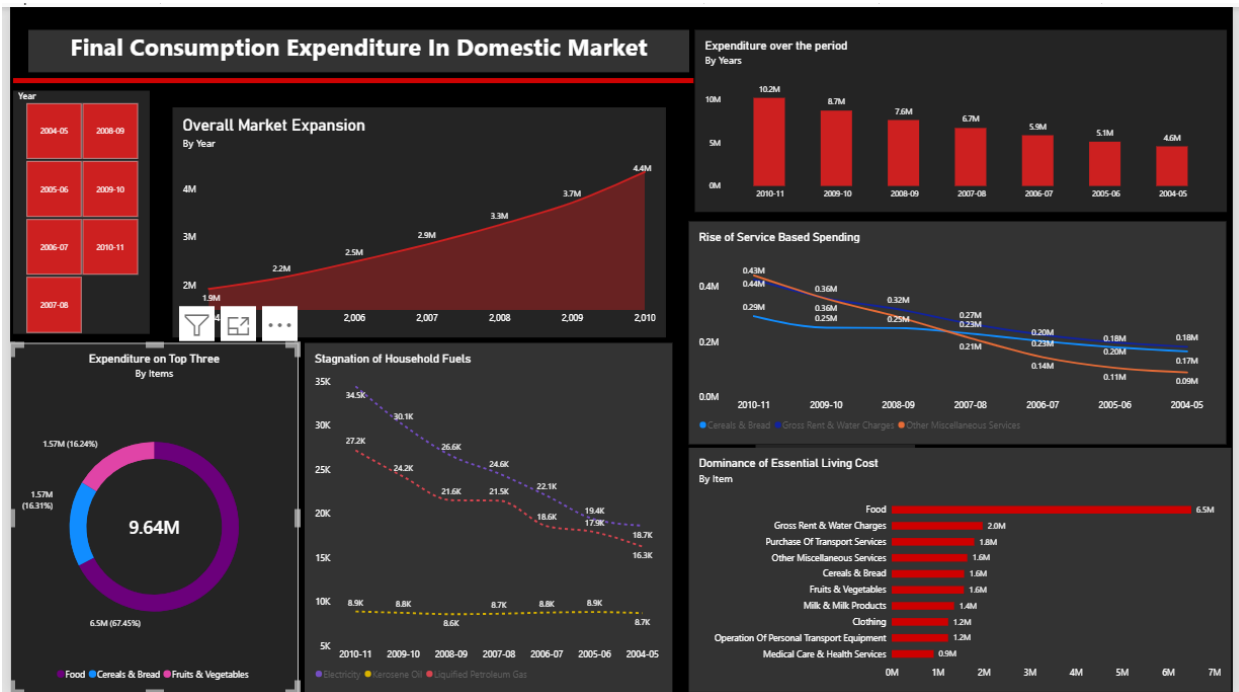
- **Data Harmonization & Engineering:** Imported raw structures into Microsoft Excel. Eliminated white spaces, synchronized nomenclature variations, and handled structural anomalies. Reconciled nested summary rows to guarantee that transaction items independently rolled up cleanly without double-counting overheads.
- **Baseline Aggregate Validation:** Constructed foundational Excel formulas and pivot tracking matrices to establish mathematical baselines for total macroeconomic spending across all operational categories for each fiscal year.

- **Power BI Modeling:** Loaded the sanitized consumption dataset into the Power BI ecosystem. Configured structural taxonomy tables, verified data type mappings, and validated that year-over-year profiles linked correctly.
- **Dashboard Interface Architecture:** Designed an intuitive dashboard interface utilizing card visual containers for primary KPIs, comparative bar graphs for cross-industry spending scales, stacked line visuals for macro trends, and targeted slicer controls for granular slice-and-dice exploration.

5. Key Insights

- **Exponential Macro Expansion:** Total verified private consumption expenditure grew aggressively, starting from an initial baseline of 1,925,592 in the 2004-05 cycle and compounding up to 4,359,792 by the 2010-11 cycle. This demonstrates a massive overall market scale expansion of 126.4%.
- **Category Market Dominance:** The 'Food, Beverages & Tobacco' segment firmly established itself as the leading economic consumer engine. Standing at an unparalleled 1,594,534 during the 2010-11 closing window, it captured a significant 36.6% market share of the total tracked national expenditure portfolio.
- **Top-Tier Micro Expenditure Drivers:** The highest individual consumer items driving the 2010-11 volume were led by 'Other Miscellaneous Services' at 440,503, followed closely by structural 'Gross Rent & Water Charges' at 425,533, and macro logistics via 'Purchase of Transport Services' marking 357,111.
- **Accelerated Dynamic Sectors:** The 'Miscellaneous Goods & Services' group showed the highest momentum over the timeline, surging by a remarkable 285.3% increase from 172,082 to 663,117.

6. Screenshots



7. Files Included

- "Consumption Source.xlsx" – Cleaned Data basic analysis
- "Dashboard.pbix" - Power BI dashboard
- 'README.md' – Project Description

8. How to Use

- **Step 1: Data Architecture Review:** Open the main Excel workbook file to review raw source definitions, verified staging fields, and the structural integrity of baseline equations.
- **Step 2: Analytical Visual Exploration:** Launch the master dashboard file via Power BI Desktop. Click through interactive category filters, hover over historical trend graphs, and explore cross-year macro shifts using built-in interactive slicers.